Framing the Truth: Flipping Black-Box LLM Answers via Semantically-Equivalent Paraphrasing

Noam Cohen*
Tel Aviv University

ABSTRACT

Large language models (LLMs) are increasingly trusted as authorities for knowledge-intensive tasks such as fact checking. We ask whether this trust is robust to *framing*: can a subtle, meaning-preserving rephrasing of a factual question—one that a human would accept as the same question, with no typos and no invisible characters—systematically steer a black-box LLM toward a factually *wrong* answer? We propose a black-box attack that couples an auxiliary LLM, used to propose natural paraphrases, with an evolutionary search that selects and mutates the rephrasings most effective at flipping the target model's answer. A central component is a semantic-equivalence validator (a natural-language-inference model plus an LLM judge, spot-checked by humans) that keeps the attack honest by rejecting any rephrasing that changes the question's meaning. We plan to measure attack success rate across several commercial and open black-box LLMs on factual question-answering benchmarks, compare against naive-rephrasing baselines, study cross-model transferability, and characterize which framing patterns are most potent. Demonstrating such rephrasings would quantify LLM sensitivity to framing effects and caution against relying on LLMs as impartial arbiters of fact.

1 INTRODUCTION

LLMs are rapidly being adopted as trusted oracles: users, journalists, and automated pipelines increasingly pose factual questions to them and treat the answers as authoritative, including for fact checking and retrieval-augmented question answering. This trust rests on an implicit assumption of *robustness*—that the answer depends on the *content* of a question, not on its incidental *phrasing*.

We investigate whether that assumption holds. Concretely, given a factual question q with an unambiguous ground-truth answer y, we seek a rephrasing q' such that q' is semantically equivalent to q yet the target model M answers incorrectly, i.e., $\text{equiv}(q, q') \wedge M(q') \neq y$, with high probability over the model's sampling. We take the example "Who won the 2024 US presidential election?" as a running illustration. Crucially, we restrict the adversary to *natural* rephrasings: no typos, no invisible or non-standard Unicode, and no adversarial suffixes. The rephrasing must be one a human reader would deem to ask the very same question. Under this constraint, a successful attack is not a "gotcha" but evidence of a genuine *framing effect*—a sensitivity to presentation long documented in human decision making [9], here exhibited by a system we treat as objective.

Our threat model is deliberately weak and therefore realistic: the adversary has only black-box query access to the target model and cannot observe gradients, logits beyond what the API exposes, or internal states. This mirrors how commercial LLMs are actually accessed. The high-level approach is an iterative optimizer: a strong auxiliary LLM proposes candidate rephrasings, each candidate is filtered by a semantic-equivalence validator, surviving candidates are scored by how strongly the target model is pushed toward a wrong answer, and an evolutionary loop mutates and recombines the best candidates over successive generations.

Our intended contributions are: (i) a formulation of *framing robustness* for factual QA and an accompanying honesty constraint that makes the attack meaningful; (ii) a black-box attack combining auxiliary-LLM paraphrase generation with evolutionary search under a semantic-equivalence guard; and (iii) an empirical study of attack success and transferability across multiple LLMs, together with a qualitative taxonomy of the framing patterns that most reliably induce errors.

2 RELATED WORK

Adversarial attacks on LLMs. A large body of work optimizes adversarial triggers appended to a prompt, such as the gradient-based GCG attack [11] and its projected-gradient and REINFORCE variants [3, 4], or generates stealthier jailbreak prompts as in AutoDAN [7] and Best-of-N [5]. These methods overwhelmingly target *safety alignment*—eliciting harmful content—and typically produce text that is unnatural (adversarial suffixes) or relies on perturbations a human would flag. Our goal differs on both axes: we target factual *correctness* rather than harmfulness, and we constrain perturbations to fluent, semantically equivalent rephrasings.

Manipulating perceived authority. Closest to our aim, recent work laundering AI authority with adversarial examples [10] studies how adversarial inputs can co-opt an LLM's perceived credibility. That work relies on optimized adversarial perturbations to *borrow* the model's authority for a chosen claim; we instead *undermine* correctness on questions the model would otherwise answer rightly, using only fluent, semantically equivalent rephrasings and no unnatural perturbation. This isolates the framing effect itself, and keeps the attack surface identical to how ordinary users legitimately query the model.

Prompt sensitivity and framing. Framing effects are a classical finding in human judgment: logically equivalent presentations of a choice yield systematically different responses [9]. A parallel brittleness is documented in LLMs: their factual answers are inconsistent across paraphrased queries [2], and their outputs are highly sensitive to superficial prompt formatting [8], while benchmarks such as TruthfulQA [6] probe truthfulness under suggestive questions. Crucially, these works *measure* sensitivity over naturally occurring or randomly varied phrasings; we instead *search* for a worst-case, semantically equivalent rephrasing that maximizes the probability of a factual error, turning a documented statistical brittleness into a directed, black-box attack.

*ID 208012252.

Prompt optimization. Our search adapts reflective prompt-evolution methods such as GEPA [1] to generate and refine candidate rephrasings.

3 WORK PLAN

We target the final report by September 10 and structure the work into four sequential phases.

Phase 0 — Setup and baselines. Select target black-box models spanning vendors and scales (e.g., a hosted GPT-class model and an open Llama-class model served behind an API). Curate a factual QA set with unambiguous, verifiable ground truth: a filtered subset of an established open-domain QA benchmark (e.g., TriviaQA or Natural Questions, restricted to items with a single canonical answer) augmented with a small hand-built set of temporally unambiguous facts (e.g., recent election and sports outcomes) to stress-test recency. Define the semantic-equivalence criterion operationally and build the validator: a natural-language-inference model checking bidirectional entailment plus an LLM judge, calibrated against human annotations on a sample. Establish baselines: the models' native error rate, and error-rate variance under *naive* paraphrases (random meaning-preserving rewrites), which bounds the effect attributable to noise rather than to optimization.

Phase 1 — Attack loop. Implement the core optimizer. An auxiliary LLM generates candidate rephrasings conditioned on the question and on high-scoring ancestors; each candidate passes through the equivalence validator; survivors are queried against the target and scored by a wrongness signal. Because commercial black-box APIs often do not expose reliable token probabilities, our primary signal is the empirical frequency of an incorrect answer under repeated sampling (using token log-probabilities only when the API exposes them), extracted with a robust answer-normalization protocol. An evolutionary controller performs selection, LLM-guided mutation, and recombination across generations. We will compare this against a greedy / best-of-N ablation and, optionally, an agentic optimizer [1].

Phase 2 — Evaluation and analysis. Measure attack success rate (fraction of questions for which a *valid* rephrasing flips the answer) per model and dataset, with the equivalence constraint enforced and human-verified on a subsample. Study transferability: do rephrasings optimized on one model fool another? Run ablations on the equivalence guard, the search strategy, and the choice of auxiliary model. Qualitatively cluster successful rephrasings into a taxonomy of framing patterns (e.g., presupposition, leading phrasing, false dichotomy, anchoring).

Phase 3 — Defenses and write-up. Discuss and lightly prototype defenses (self-consistency over paraphrases, answer aggregation, framing-robust prompting) and report their effect on attack success. Finalize the report and release documented, reproducible code.

Risks and mitigations. The chief risk is guaranteeing true semantic equivalence; we mitigate it with a conservative multi-signal validator and human spot-checks, and we report attack success only over human-verified valid rephrasings. API cost is controlled by favoring small/open models and caching. Model hedging or refusals are handled by a predefined answer-extraction and abstention protocol.

4 ETHICAL CONSIDERATIONS

Our study demonstrates how to induce factual errors in systems the public treats as authoritative, and our running examples touch on politically sensitive facts; we therefore proceed defensively. All experiments target our own queries against publicly accessible models—we neither publish nor amplify any misleading content, and any wrong answers we elicit concern facts with well-established ground truth used solely for measurement. We will disclose potent framing patterns responsibly to the relevant model providers before public release, and we frame the project as motivating *defenses*: Phase 3 prototypes mitigations (paraphrase self-consistency, answer aggregation) so that the takeaway is improved robustness rather than a reusable exploit. We will comply with each provider's terms of service and rate limits throughout.

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